

Think of the MACD as a car's dashboard showing both speed (momentum) and direction (trend). The relationship between its lines tells you if the market is accelerating, decelerating, or potentially changing direction.

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1. def calculate_macd(data, fast_period=12, slow_period=26, signal_period=9):
2.     # Calculate fast and slow EMAs
3.     ema_fast = data.ewm(span=fast_period, adjust=False).mean()
4.     ema_slow = data.ewm(span=slow_period, adjust=False).mean()
5.
6.     # Calculate MACD line
7.     macd_line = ema_fast - ema_slow
8.
9.     # Calculate signal line
10.    signal_line = macd_line.ewm(span=signal_period, adjust=False).mean()
11.
12.    # Calculate histogram
13.    histogram = macd_line - signal_line
14.
15.    return macd_line, signal_line, histogram
16.
17. # Calculate MACD
18. apple['MACD'], apple['MACD_Signal'], apple['MACD_Histogram'] =
calculate_macd(apple['Close'])
```

